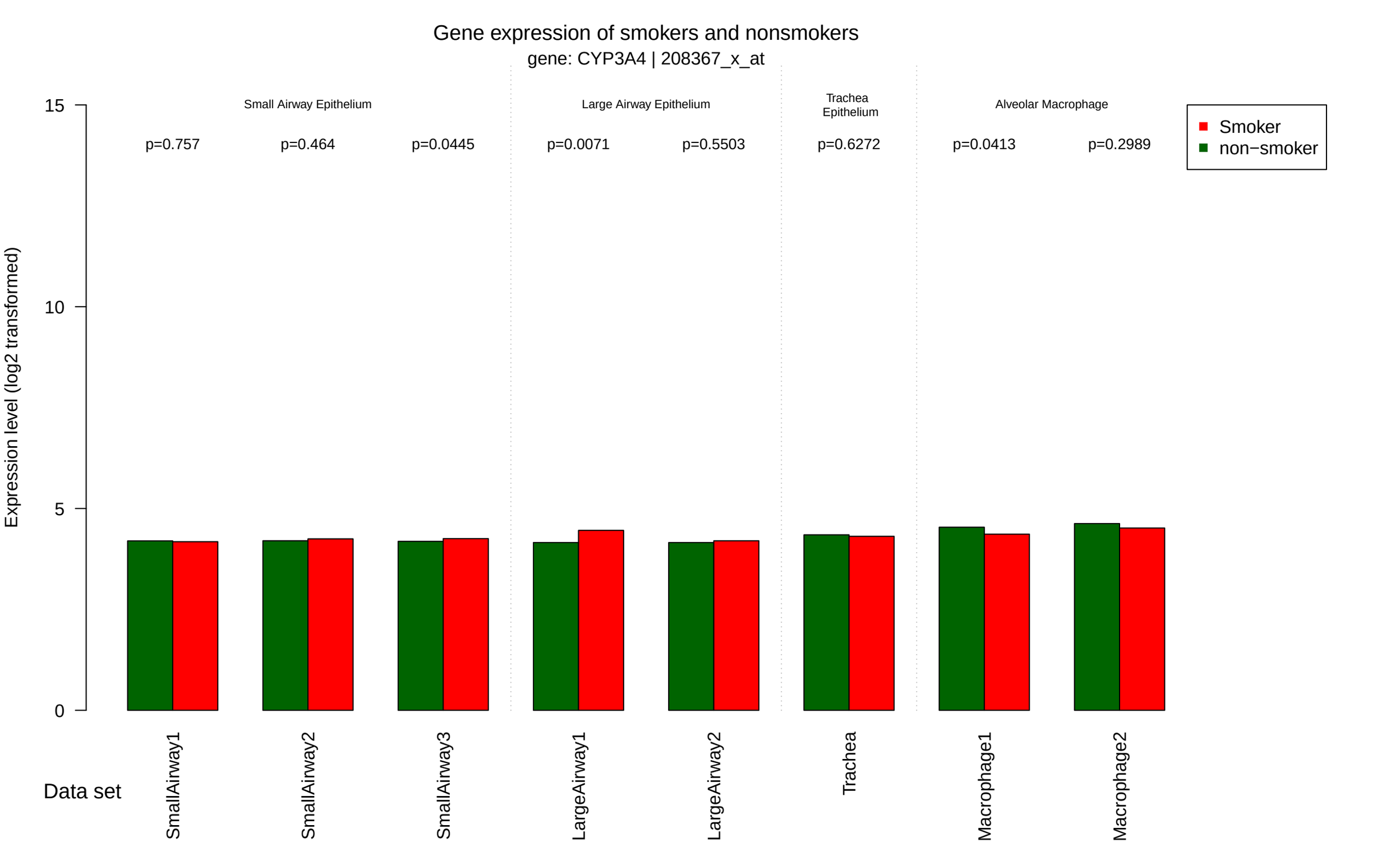
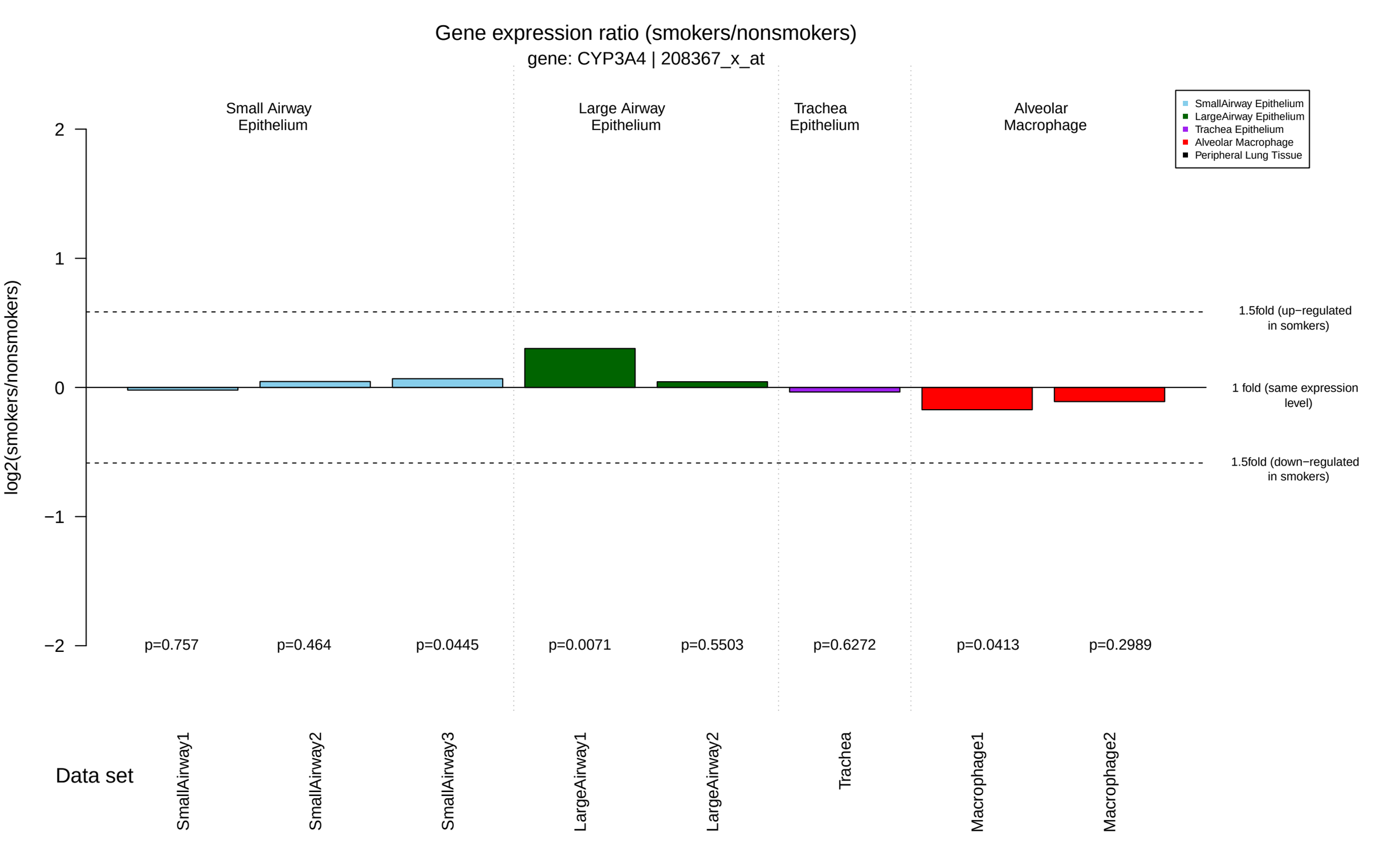
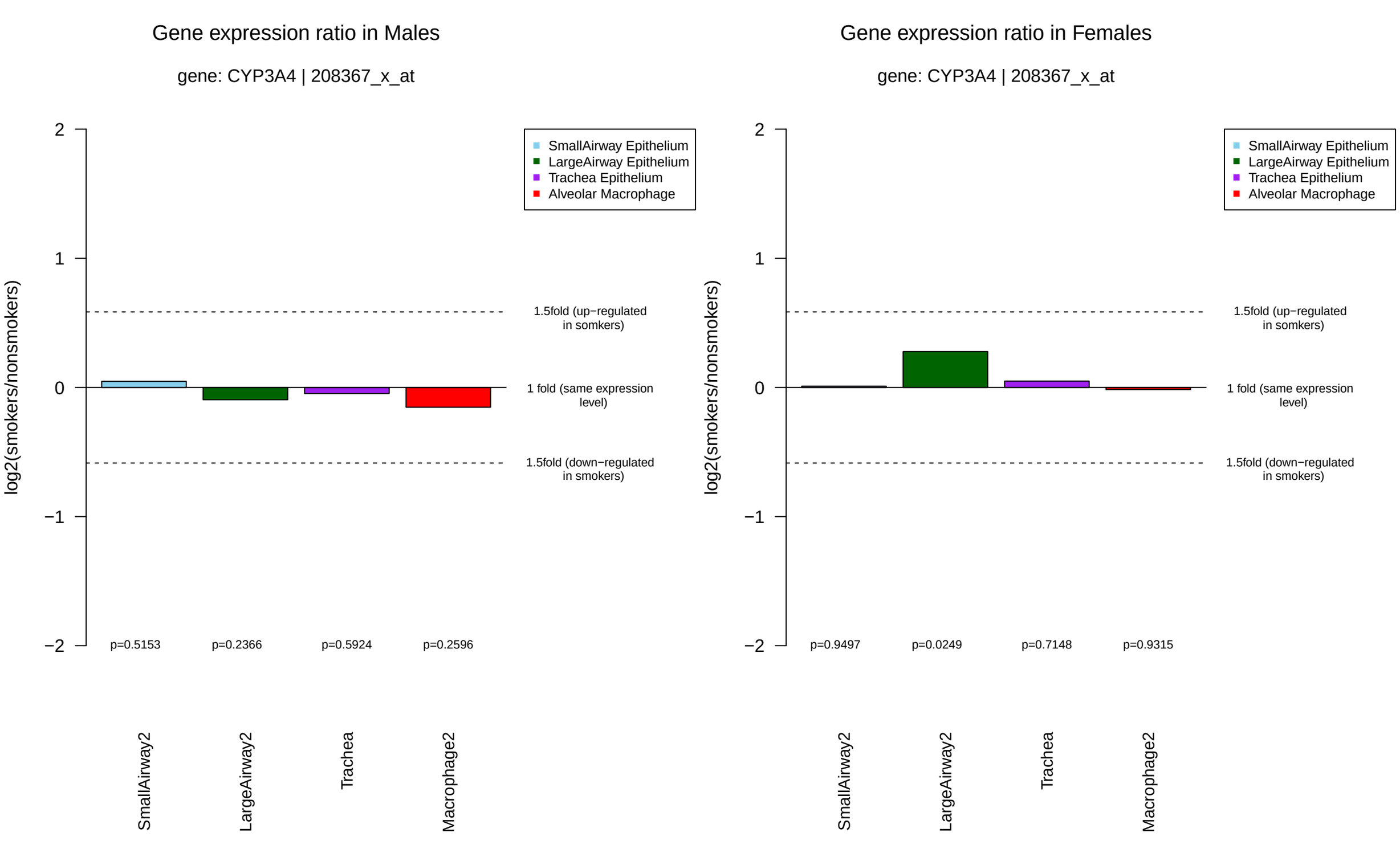
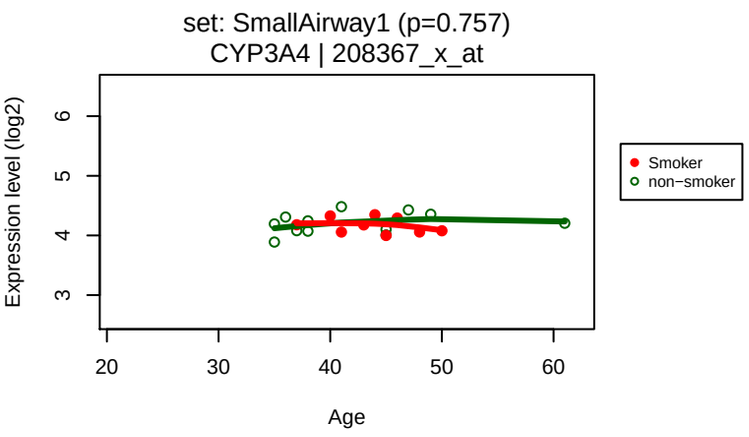
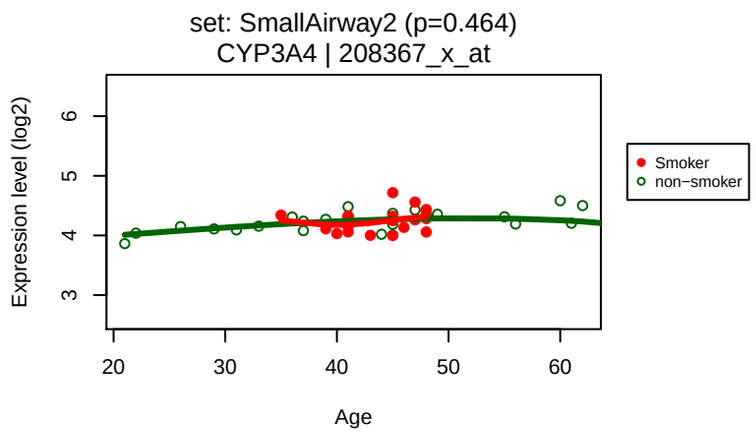
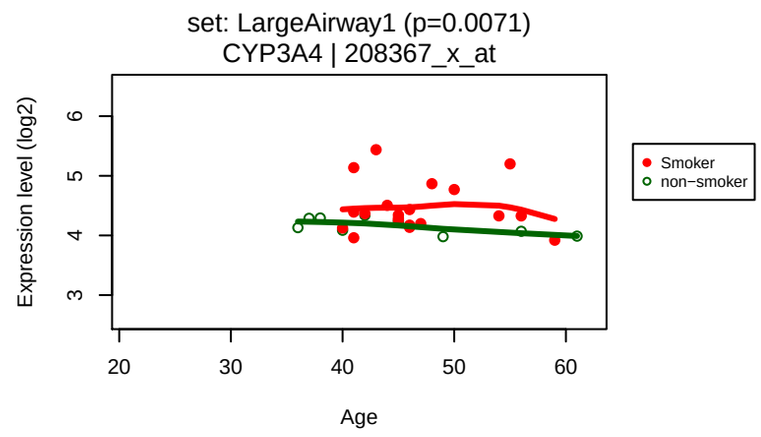

Gene expression of smokers and nonsmokers
gene: CYP3A4 | 208367_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.757
p=0.464
p=0.0445
p=0.0071
p=0.5503
p=0.6272
p=0.0413
p=0.2989
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

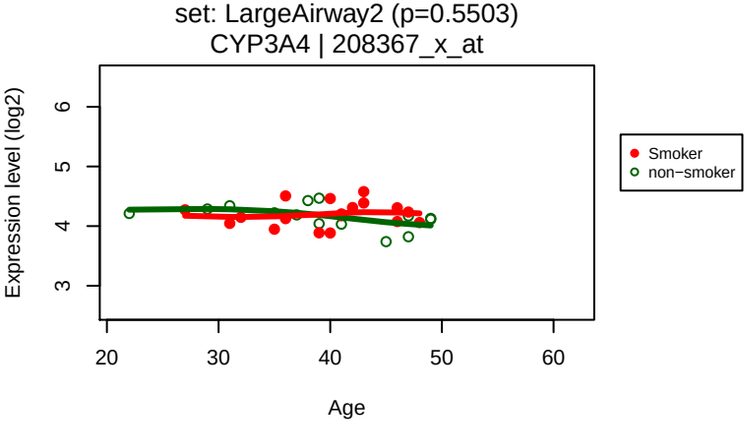

Gene expression ratio (smokers/nonsmokers)
gene: CYP3A4 | 208367_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.757
p=0.464
p=0.0445
p=0.0071
p=0.5503
p=0.6272
p=0.0413
p=0.2989
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

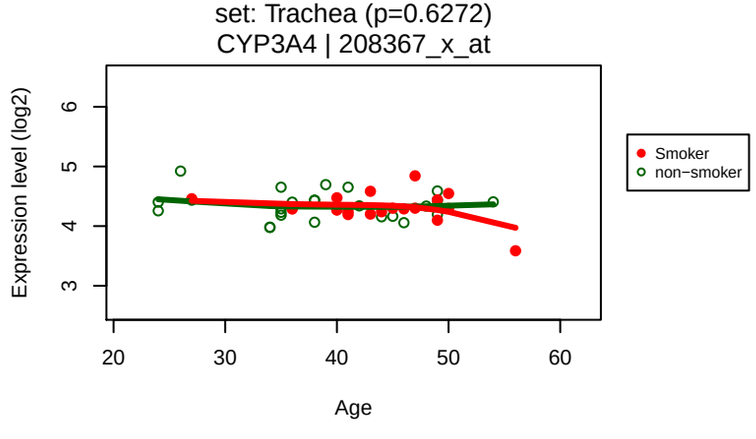

Gene expression ratio in Males
gene: CYP3A4 | 208367_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5153
p=0.2366
p=0.5924
p=0.2596
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CYP3A4 | 208367_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9497
p=0.0249
p=0.7148
p=0.9315
SmallAirway2
LargeAirway2
Trachea
Macrophage2

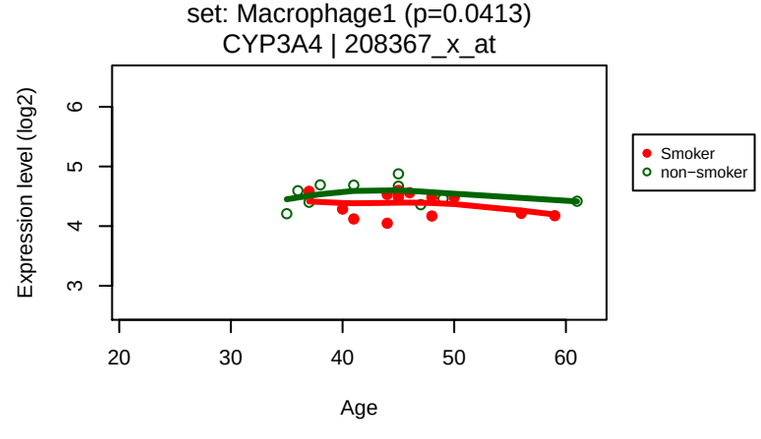

set: SmallAirway1 (p=0.757)
CYP3A4 | 208367_x_at
Expression level (log2)
Age
Smoker
non−smoker

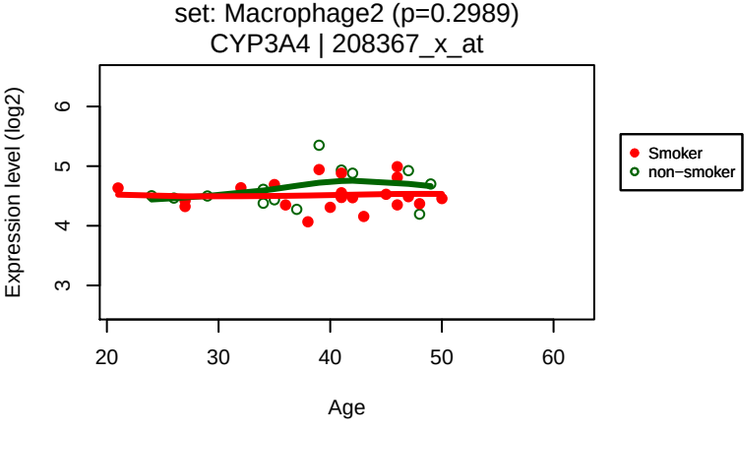

set: SmallAirway2 (p=0.464)
CYP3A4 | 208367_x_at
Expression level (log2)
Age
Smoker
non−smoker

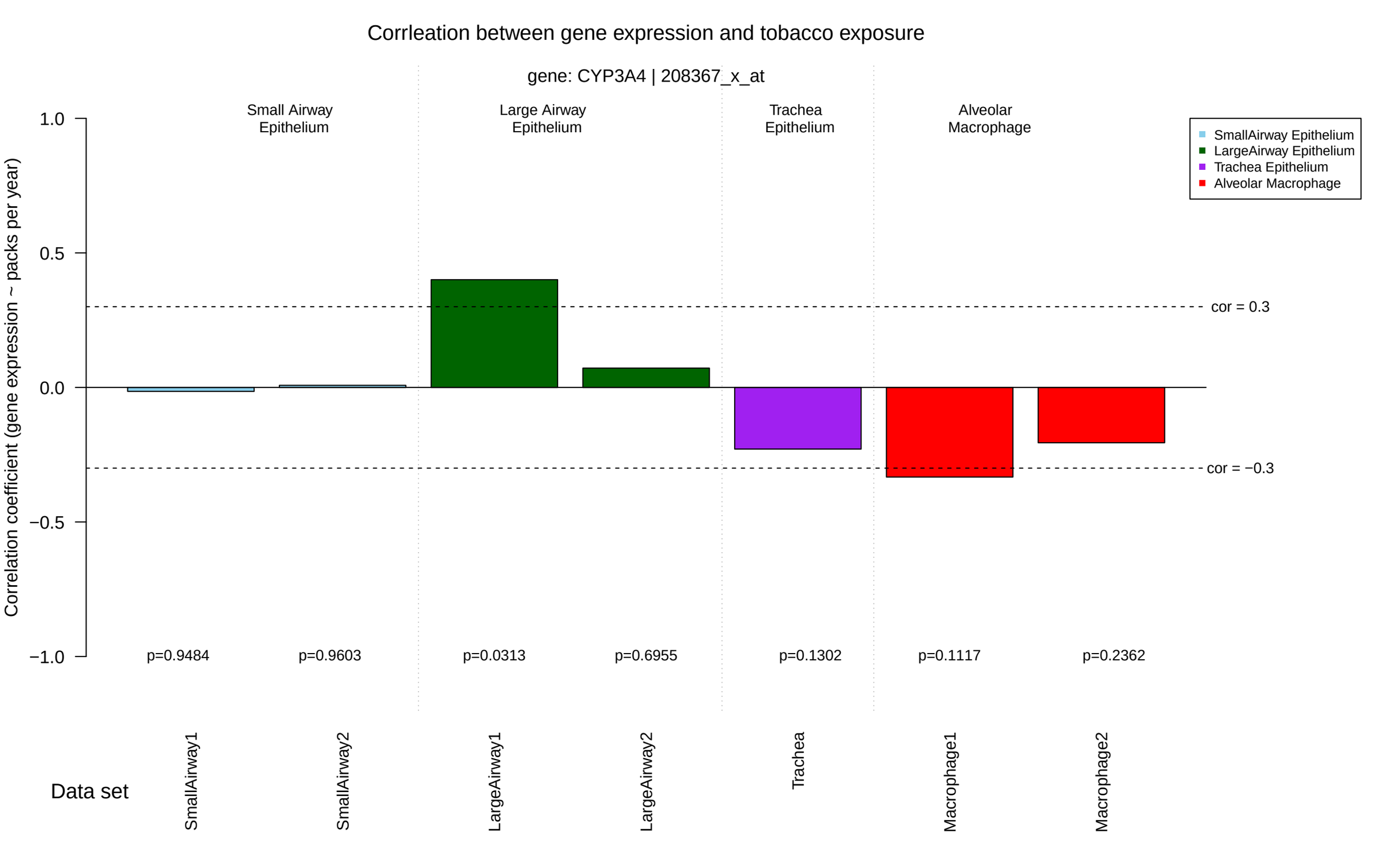

set: LargeAirway1 (p=0.0071)
CYP3A4 | 208367_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5503)
CYP3A4 | 208367_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6272)
CYP3A4 | 208367_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0413)
CYP3A4 | 208367_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2989)
CYP3A4 | 208367_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CYP3A4 | 208367_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9484
p=0.9603
p=0.0313
p=0.6955
p=0.1302
p=0.1117
p=0.2362
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2